1. Motivation and Research Question

What am I studying? I am using multiple linear regression in SAS to examine how healthy life expectancy, freedom, generosity, and perceptions of corruption relate to national happiness scores in the World Happiness Report 2024.
Research question: How are a country's healthy life expectancy, freedom to make life choices, generosity, and perceptions of corruption associated with its overall happiness score?
Why is this question interesting? Understanding the factors associated with national happiness can help identify aspects of well-being that may contribute to a country's quality of life. The findings may provide insights for policymakers seeking to improvepeople's well-being, although the analysis can identify associations rather than establish causation.

2. Data Description

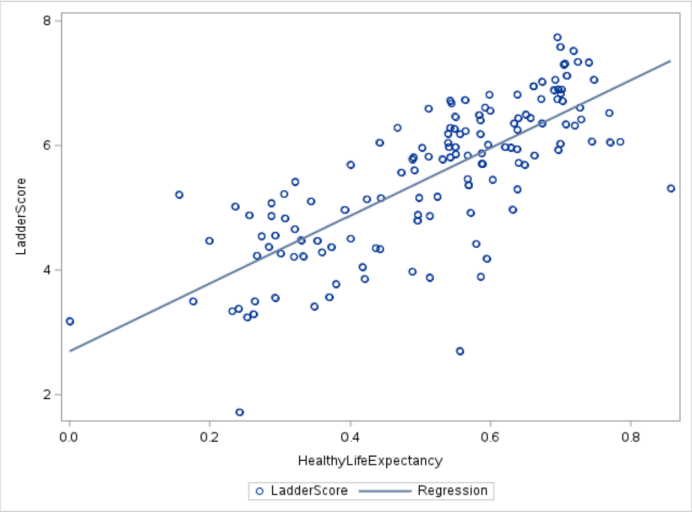
Source: Data is taken indirectly from the “World Happiness Report 2024” published by the Wellbeing Research Centre at the University of Oxford through Kaggle (https://www.kaggle.com/datasets/jainaru/world-happiness-report-2024-yearly-updated/data)
Number of observations: 140 observations, each representing a country
Response variable: Happiness Score (Ladder score) - a national average of survey respondents' self-rated life satisfaction on a 0–10 scale
Predictor variables:

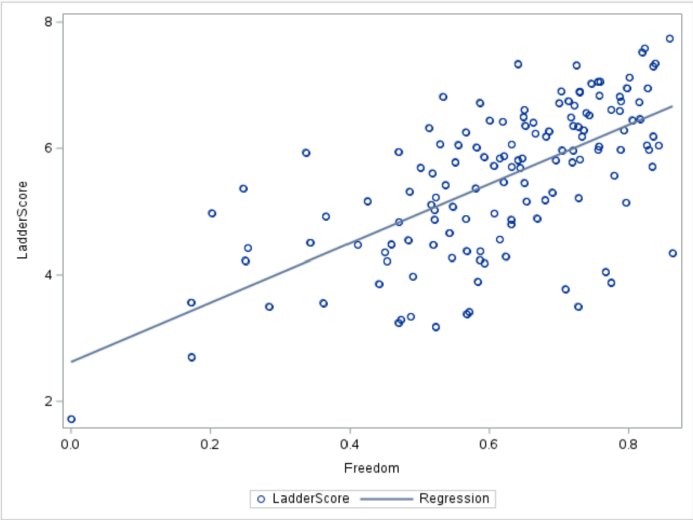
- Healthy life expectancy: The calculated contribution of healthy life expectancy to the happiness score
- Freedom to make life choices: The calculated contribution of freedom to the happiness score
- Generosity: The calculated contribution of generosity to the happiness score, based on charitable donation data adjusted for GDP per capita.
- Perceptions of corruption: The calculated contribution of perceptions of corruption to the happiness score

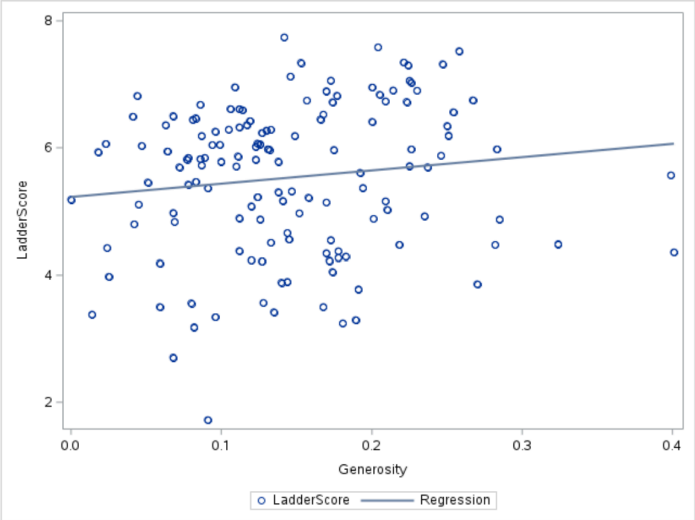
3. Data exploration

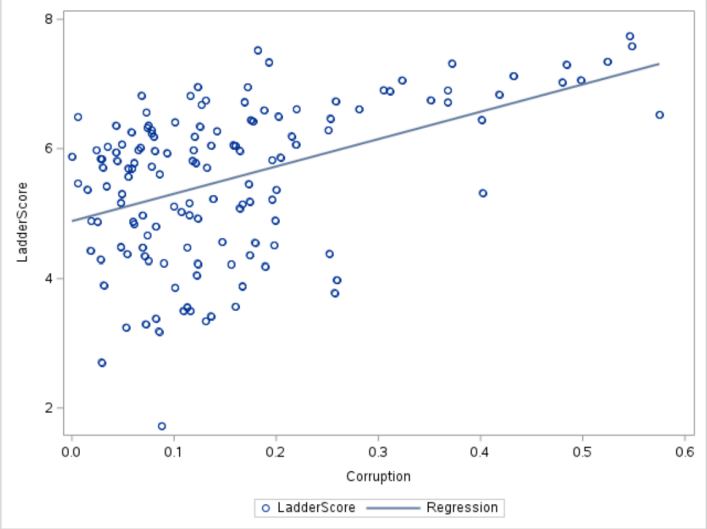
Scatterplots of each predictor vs. happiness score:

- Healthy life expectancy: shows a strong positive relationship with happiness. Countries with higher healthy life expectancy contributions tend to have higher happiness scores.
- Freedom: shows a positive relationship with happiness, suggesting that countries with higher freedom contributions tend to have higher happiness scores.
- Generosity: shows a relatively weak relationship with happiness.
- Perceptions of corruption: shows a moderate positive relationship with happiness









4. Regression Modeling and Analysis

(1) Initial Model and Evaluation

Model equation:
LadderScore-hat = B0 + B1(HealthLifeExpectancy) + B2(Freedom) + B3(Genoriosity) + B4(Corruption) + Ei
LadderScore-hat = 1.499 + 4.098(HealthLifeExpectancy) + 2.753(Freedom) + 0.413Generosity) + 0.830(Corruption)
Overall F-test:
Ho: B1 = B2 = B3 = B4 = 0 Ha: at least one slope is non-zero
Since p-value<0.05, we reject the null hypothesis and conclude that there is sufficient evidence that at least one predictor is associated with the happiness score.

The model has an **R^2** of 0.7221 and an **adjusted R^2** of 0.7139. This means the model explains approximately 72.2% of the variation in happiness scores.
Individual p-values: HealthLifeExpectancy and Freedom both have p-values<.0001, meaning they both have significant associations with happiness scores after controlling for the other predictors. Meanwhile, there is insufficient evidence to conclude that Generosity and Corruption have an individual association with happiness after accounting for the other variables.

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	4	140.05931	35.01483	87.71	<.0001
Error	135	53.89353	0.39921		
Corrected Total	139	193.95284			

Root MSE	0.63183	R-Square	0.7221
Dependent Mean	5.53089	Adj R-Sq	0.7139
Coeff Var	11.42368		

Parameter Estimates								
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Variance Inflation	95% Confidence Limits	
Intercept	1	1.49940	0.24713	6.07	<.0001	0	1.01066	1.98814
HealthyLifeExpectancy	1	4.09832	0.37553	10.91	<.0001	1.33558	3.35563	4.84101
Freedom	1	2.75260	0.37741	7.29	<.0001	1.30952	2.00619	3.49901
Generosity	1	0.41314	0.75948	0.54	0.5874	1.08325	-1.08888	1.91517
Corruption	1	0.83040	0.47922	1.73	0.0854	1.27428	-0.11735	1.77815

(2) Diagnostics and Modeling Decisions

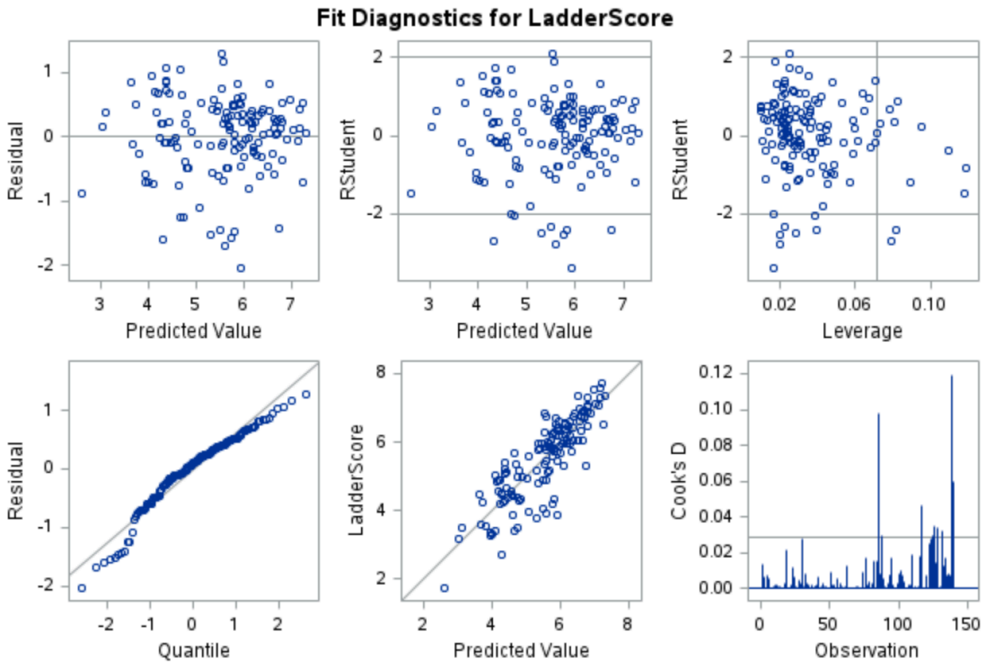
Multicollinearity: all 4 predictors have VIF close to 1, so we can say that they have little overlap with each other

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Checking assumptions:

- Linearity: the residual plot shows no clear curve or pattern, suggesting that the linearity assumption is reasonably satisfied.
- Equal variance: The residuals appear to have a relatively constant spread, although there are several larger residuals at lower predicted values. This suggests that the equal variance assumption may be slightly violated.

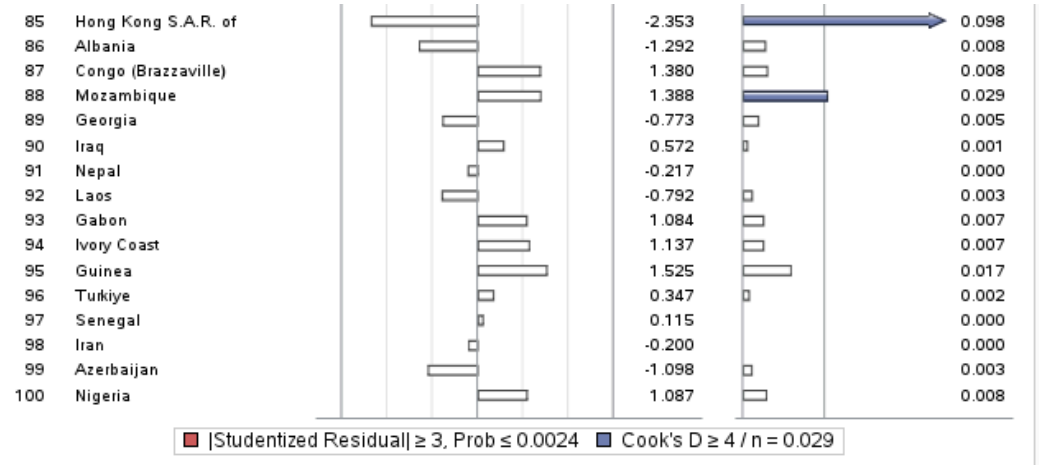
- Normality of residuals: The Q-Q plot shows small deviations from the diagonal line at both tails, suggesting that the residuals are not perfectly normally distributed, particularly in the lower tail. However, we can still say the normality assumption is reasonably satisfied.

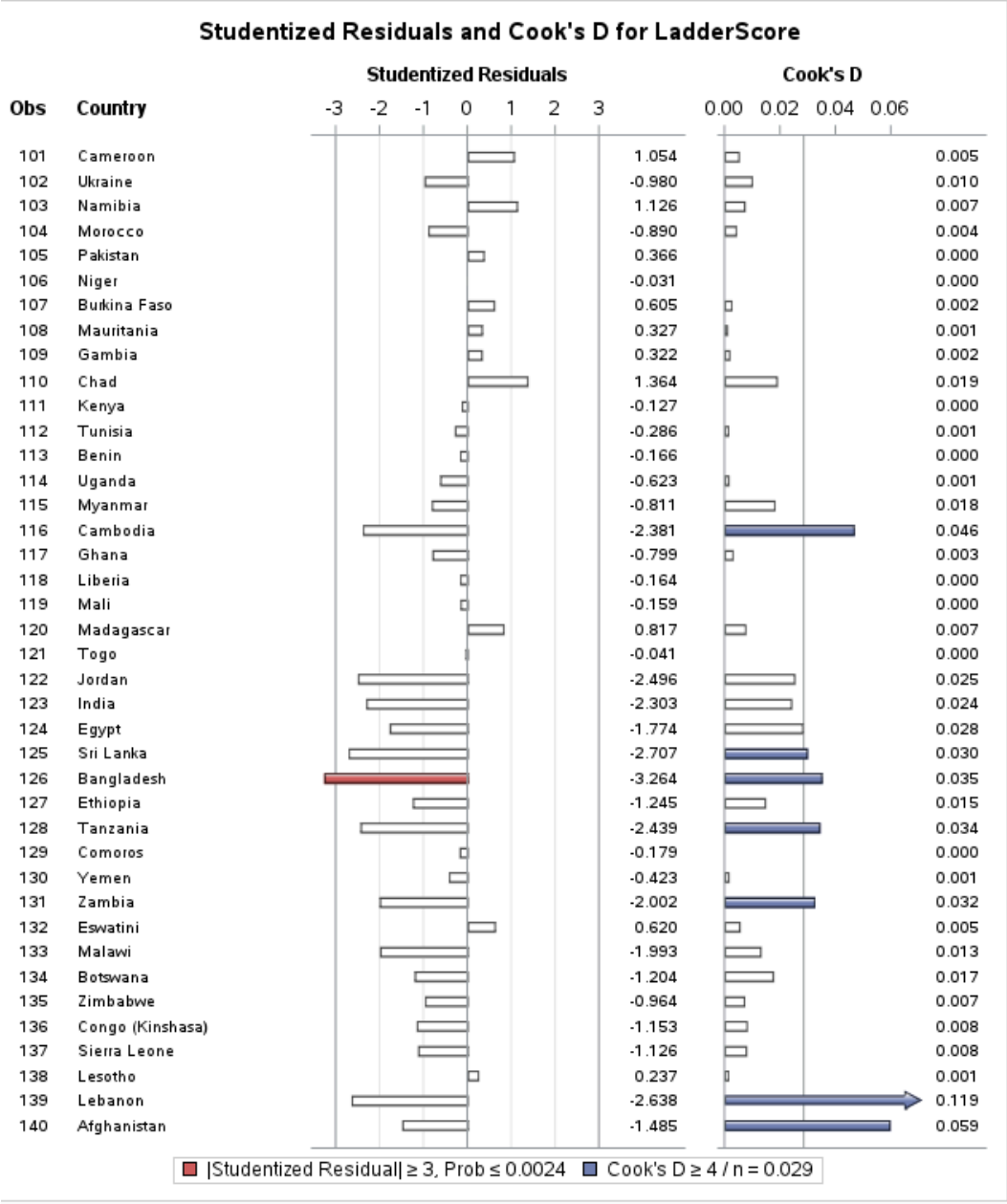


Unusual observations:

- Large studentized residuals: Bangladesh (−3.264) has a studentized residual below −3, suggesting its happiness score is substantially lower than predicted by the model.
- High Cook's distance: Lebanon (0.119), Hong Kong (0.098), and Afghanistan (0.059) all largely exceed the reference line. These countries may have a strong influence on the regression estimates.
- Other potentially influential observations: Cambodia (0.046), Tanzania (0.034), Zambia (0.032), and Sri Lanka (0.030) also exceed or are close to the Cook's distance reference line but not much.

=> My decision is that I would retain these observations in the initial model because they may represent genuine differences between countries rather than data errors.





Variable selection:

Summary of Forward Selection							
Step	Variable Entered	Number Vars In	Partial R-Square	Model R-Square	C(p)	F Value	Pr > F
1	HealthyLifeExpectancy	1	0.5771	0.5771	69.4703	188.30	<.0001
2	Freedom	2	0.1376	0.7147	4.6326	66.05	<.0001

Summary of Backward Elimination							
Step	Variable Removed	Number Vars In	Partial R-Square	Model R-Square	C(p)	F Value	Pr > F
1	Generosity	3	0.0006	0.7215	3.2959	0.30	0.5874
2	Corruption	2	0.0069	0.7147	4.6326	3.35	0.0692

Summary of Stepwise Selection								
Step	Variable Entered	Variable Removed	Number Vars In	Partial R-Square	Model R-Square	C(p)	F Value	Pr > F
1	HealthyLifeExpectancy		1	0.5771	0.5771	69.4703	188.30	<.0001
2	Freedom		2	0.1376	0.7147	4.6326	66.05	<.0001

All 3 selection methods retained (HealthyLifeExpectancy, Freedom) and eliminated (Generosity, Corruption)

(3) Final Model and Interpretation

Fitted regression equation:
LadderScore-hat = 1.476 + 4.278(HealthLifeExpectancy) + 2.943(Freedom)
Interpretation: Holding Freedom constant, a one-unit increase in the HealthyLifeExpectancy is associated with a 4.2778-point increase in the predicted happiness score. Holding HealthyLifeExpectancy constant, a one-unit increase in Freedom is associated with a 2.9435-point increase in the predicted happiness score.
R^2 = 0.715 and Adjusted R^2 = 0.710; this means the model explains approximately 71.5% of the variation in happiness scores.
=> Although the reduced model explains slightly less variation in happiness scores, the decrease is small. Therefore, I selected the reduced model because it is simpler and focuses on the two predictors with the strongest statistical evidence of association with happiness.
95% confidence intervals:

- Healthy life expectancy: We are 95% confident that its regression coefficient lies between 3.57222 and 4.98347.
- Freedom: We are 95% confident that its regression coefficient lies between 2.22728 and 3.65963.

Changes from the initial model: The initial model included four predictors: healthy life expectancy, freedom, generosity, and perceptions of corruption. After variable selection, generosity and perceptions of corruption were eliminated because they were not statistically significant at the 5% level.

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	138.60912	69.30456	171.56	<.0001
Error	137	55.34371	0.40397		
Corrected Total	139	193.95284			

Root MSE	0.63559	R-Square	0.7147
Dependent Mean	5.53089	Adj R-Sq	0.7105
Coeff Var	11.49155		

Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	95% Confidence Limits	
Intercept	1	1.47585	0.23333	6.33	<.0001	1.01446	1.93725
HealthyLifeExpectancy	1	4.27784	0.35684	11.99	<.0001	3.57222	4.98347
Freedom	1	2.94346	0.36218	8.13	<.0001	2.22728	3.65963

5. Final Conclusion

This study examined how healthy life expectancy, freedom to make life choices, generosity, and perceptions of corruption are associated with national happiness using data from the World Happiness Report 2024. The final multiple linear regression model retained healthy life expectancy and freedom, both of which had statistically significant positive associations with happiness ($p < 0.0001$). Their 95% confidence intervals were (3.57222, 4.98347) and (2.22728, 3.65963), respectively. The model explained approximately 71.5% of the variation in happiness scores, suggesting that these two predictors account for a substantial proportion of differences in happiness scores.

The results show that healthy life expectancy and freedom contributions are strongly associated with overall happiness scores. However, because these predictors are calculated contributions to the happiness score, their relationships with happiness are partly built into the scoring methodology. Therefore, the results should not be interpreted as independent evidence that actual life expectancy and freedom cause greater happiness.

SAS Code Appendix

```
DATA Happiness;
      INFILE "/home/u64421950/STA3064/Data/World-happiness-report-2024.csv"
DLM=',' DSD FIRSTOBS=2;
      LENGTH Country $40;
      INPUT Country $
          RegionalIndicator $
          LadderScore
          UpperWhisker
          LowerWhisker
          LogGDP
          SocialSupport
          HealthyLifeExpectancy
          Freedom
          Generosity
          Corruption
          DystopiaResidual;
RUN;

/* Scatterplot: Healthy Life Expectancy vs. Happiness */
PROC SGPLOT DATA=Happiness;
    SCATTER X=HealthyLifeExpectancy Y=LadderScore;
    REG X=HealthyLifeExpectancy Y=LadderScore;
RUN;
/* Scatterplot: Freedom vs. Happiness */
PROC SGPLOT DATA=Happiness;
```